12-W1　返回电流与磁矢势

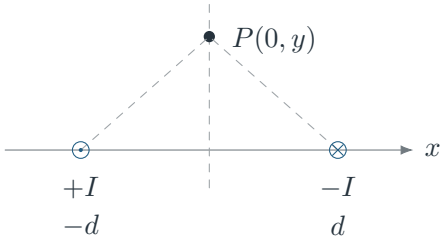
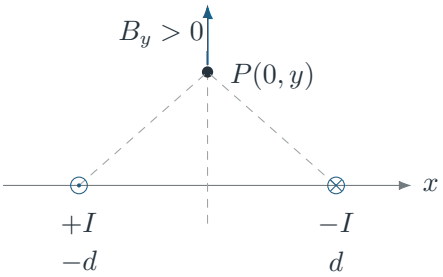


图 12-14　题图　左线电流向纸外，右线向纸内，中垂面上取场点

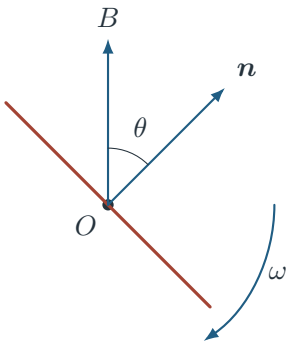


$$A_z = 0 \text{ 于 } x = 0$$

$$\partial_x A_z \neq 0$$

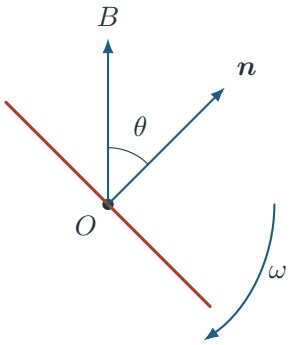
图 12-15　解答分析图　势值可相消，势的法向梯度不相消

12-W2　场强和线圈角度同时改变



$$B = B_0 e^{-t/T}$$

图 12-16　题图　转轴垂直图面，法向与场夹角改变，外场同时衰减



$$B = B_0 e^{-t/T}$$

$$\Psi = NSB\cos\theta$$

$-\dot{\Psi}$ 含两个分项

图 12-17　解答分析图　场强导数与夹角导数都进入磁通链的全导数

12-W3　匀速滑动棒的外力功率

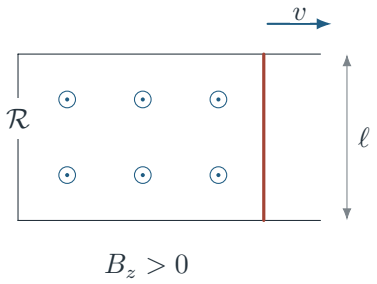


图 12-18　题图　外场向纸外，右棒向右运动，左端接电阻

$$P_{\text{ext}} = I^2\mathcal{R}$$

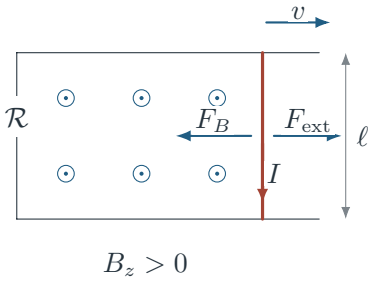


图 12-19　解答分析图　右棒电流向下，磁力向左，外力功率转为热

12-W4　两根滑动棒的相对运动

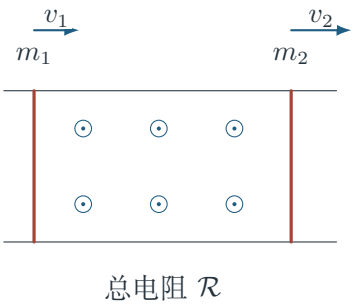


图 12-20　题图　两根棒均能移动，图示右棒比左棒快

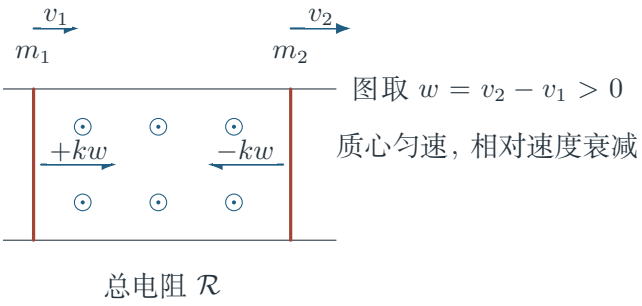


图 12-21　解答分析图　两棒受力相反，只有相对运动的动能变为热

12-W5　旋转弯导线的端电压

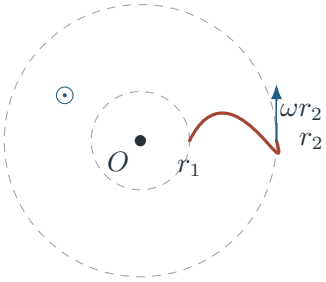
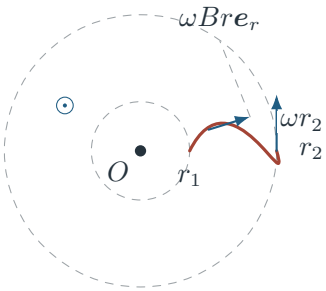


图 12-22　题图　弯导线刚性转动，内外端到轴的距离不同



$$\mathcal{E} = \int \omega B r \,\mathrm{d}r$$

$$V_2 > V_1$$

图 12-23　解答分析图　径向作用的路径积分只依赖半径平方差

12-W6　环形圆盘发电机的转矩

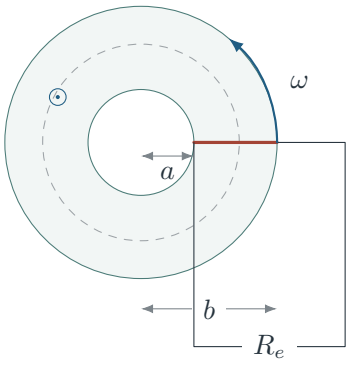
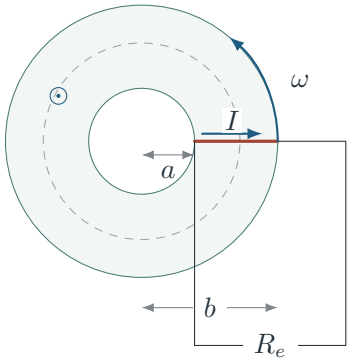


图 12-24　题图　内外圆周电刷与外电阻构成回路，盘内电流径向流动



$$R_d = \ln(b/a)/(2\pi\sigma h)$$

$$\tau_B = -IB(b^2 - a^2)/2$$

图 12-25　解答分析图　径向电流受负方位磁力，驱动转矩支付内外电阻发热

12-W7　变电流导线旁的移动线圈

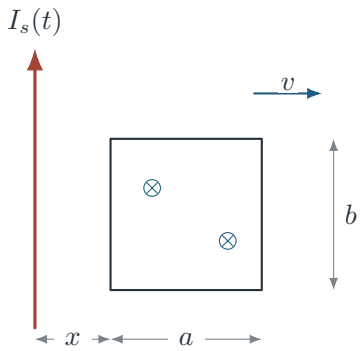


图 12-26　题图　直导线右侧磁场向纸内，矩形线圈向右离开

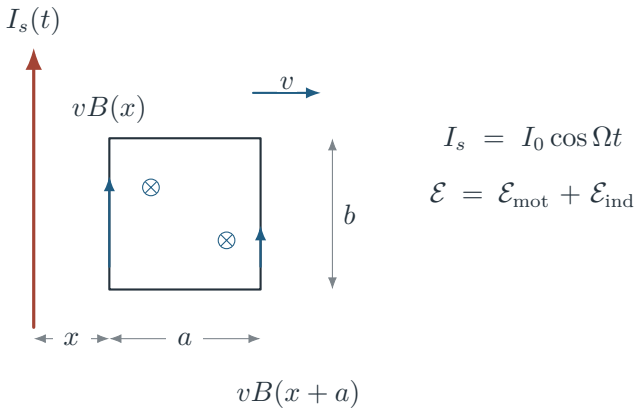


图 12-27　解答分析图　两纵向边的动生项相减，另加外场变化的感生项

12-W8　螺线管内外的感应电场

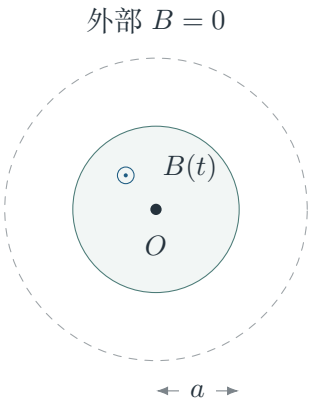
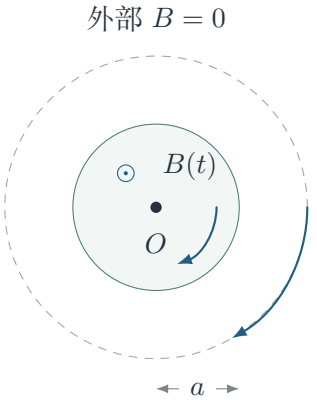


图 12-28　题图　外圆周位于磁场为零的区域，却围住变化的磁通

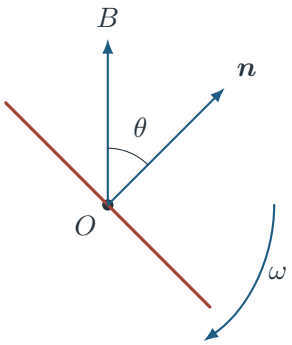


$\dot{B}>0$ 时场沿顺时针

外部 $E_\varphi \propto 1/r$

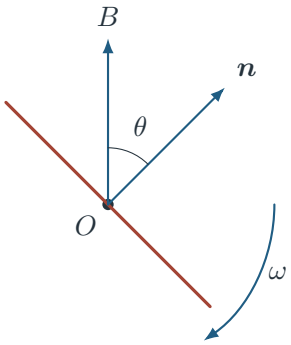
图 12-29　解答分析图　内外电场方向一致，边界连续，外部环流仍非零

12-W9　交流发电机的功率和转矩



线圈侧视

图 12-30　题图　线圈在恒定场中旋转，法向与场的夹角为转角



线圈侧视

$$I \propto \sin\theta$$

$$\tau_B \propto -\sin^2\theta$$

$$\langle P\rangle = \mathcal{E}_0^2/(2\mathcal{R})$$

图 12-31　解答分析图　电流与几何因子同时换号，磁转矩始终阻转

12-W10　两种引线走向的电压读数

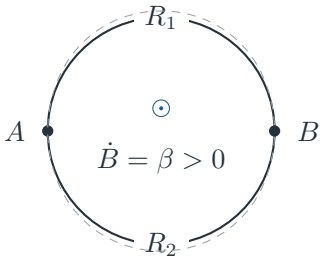


图 12-32　题图　虚线示意紧贴上、下支路的两种测量路径，正端均接右端

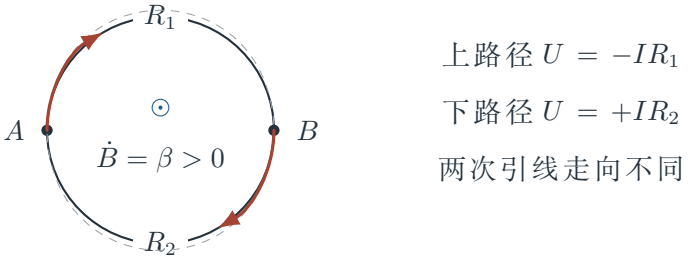


图 12-33　解答分析图　电流顺时针，上、下支路沿同一端点方向的场积分不同

12-W11　磁场梯度中的阻尼振动

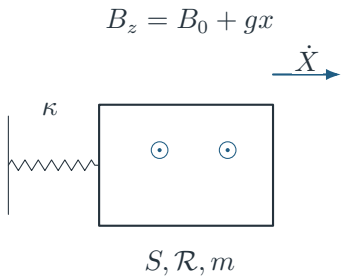
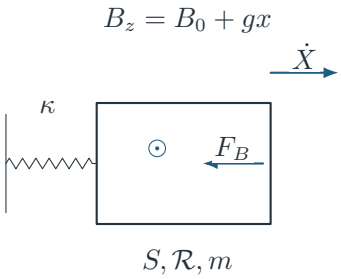


图 12-34　题图　弹簧约束线圈在具有恒定梯度的外场中平移



$$F_B = -(Sg)^2\dot{X}/\mathcal{R}$$

$$m\ddot{X} + k\dot{X} + \kappa X = 0$$

图 12-35　解答分析图　图取速度向右，磁力向左，机械能按焦耳功率损失

12-W12　薄圆盘的涡流功率

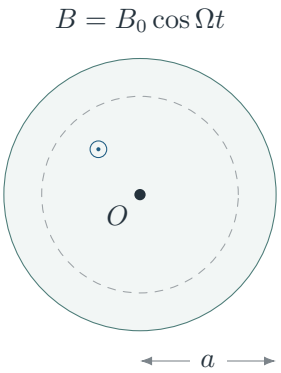
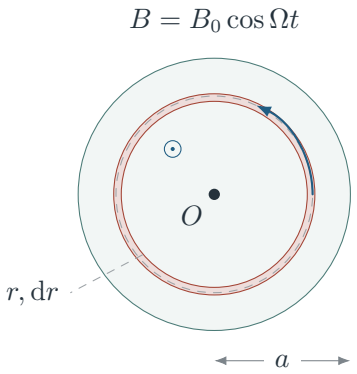


图 12-36　题图　轴对称薄圆盘受到垂直的均匀交变场



图取 $\dot{B} < 0$

$J_\varphi = \sigma E_\varphi$

环带体积 $2\pi r h \, \mathrm{d}r$

图 12-37　解答分析图　场衰减时涡流逆时针，功率由所有环带积分

12-T1　不同面积绕组的磁通链

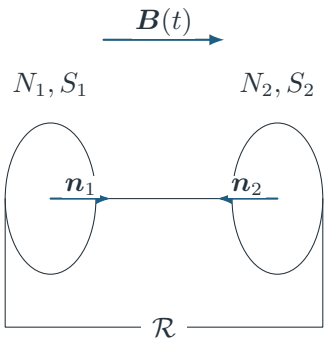
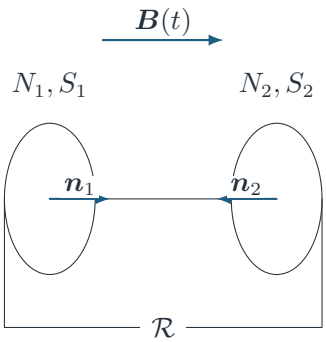


图 12-38　题图　两绕组按串联方向具有相反面积矢量



$$\Psi = (N_1S_1 - N_2S_2)B$$

反向串接使磁通链相减

图 12-56　解答分析图　总磁通链须带符号相加，局部感应场仍可存在

12-T2　膨胀圆环与衰减磁场

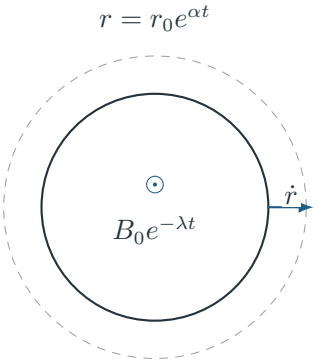
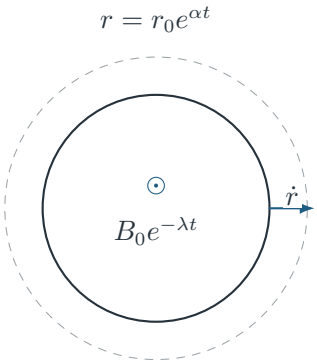


图 12-39　题图　圆环均匀扩张，向纸外的外场同时衰减



$$\mathcal{E}_{\text{mot}} = -2\alpha\Phi$$

$$\mathcal{E}_{\text{ind}} = +\lambda\Phi$$

图 12-57　解答分析图　面积变化与场变化反向贡献，可在特定速率下抵消

12-T3　弯导线平动时的电动势

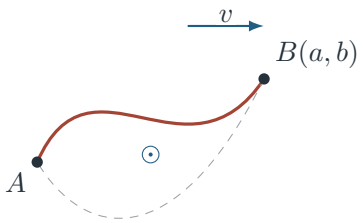


图 12-40　题图　实线为任意弯导线，虚线为返回路径

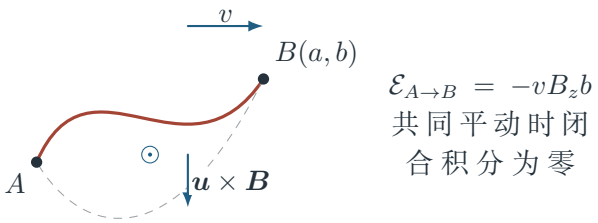


图 12-58　解答分析图　均匀平动的局部作用为常矢量，端点纵向差决定开口积分

12-T4　偏轴旋转的直导体棒

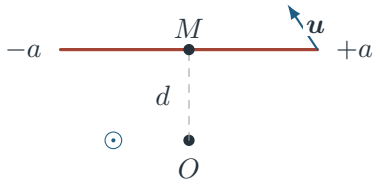
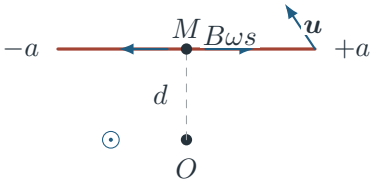


图 12-41　题图　棒不穿过转轴，两端到轴等距



$$\mathcal{E}_{-a\to+a} = 0$$

两端均比中点电势高

图 12-59　解答分析图　两半棒的动生贡献相反，沿棒存在二次电势分布

12-T5　下落导体棒的终速

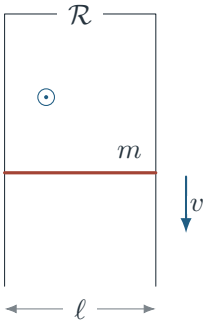
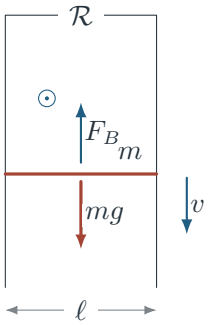


图 12-42　题图　横棒沿竖直导轨下落，顶部由电阻闭合



$$m\dot{v} = mg - kv$$

$$v_\infty = mg/k$$

$$Q = mgy - mv^2/2$$

图 12-60　解答分析图　磁力向上，随速度增大直到平衡重力

12-T6　穿过有限磁场带的线圈

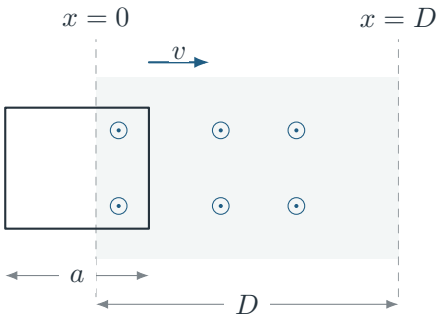


图 12-43　题图　图示进入阶段，灰绿色带内有磁场，外部为零

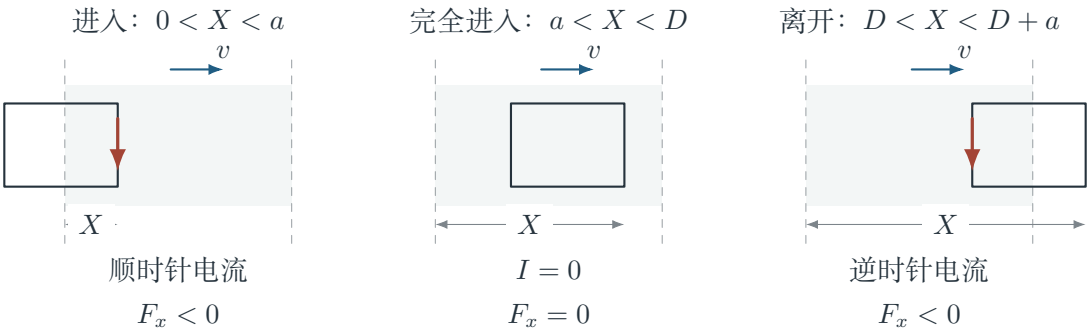


图 12-61　解答分析图　三阶段按磁场重叠面积区分，出入时电流反向而阻力均向左

12-T7　磁场变化传输的电荷

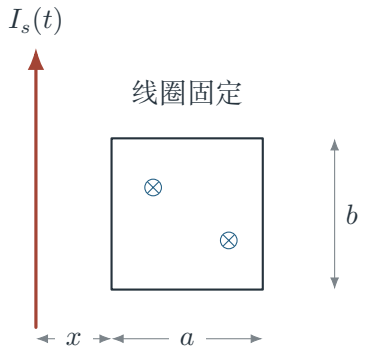


图 12-44　题图　固定矩形线圈接收直导线电流变化引起的感应

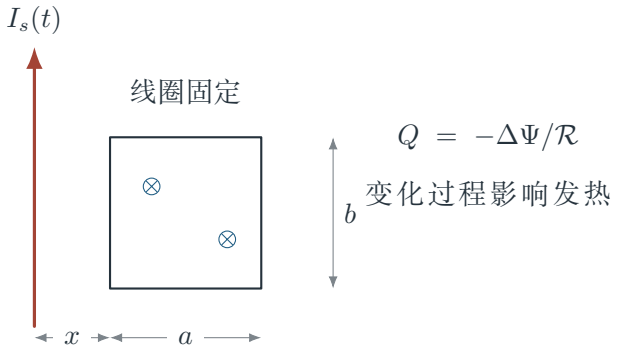


图 12-62　解答分析图　电荷由初末磁通链决定，热量由电流平方的时间积分决定

12-T8　在无磁场处扩张的圆环

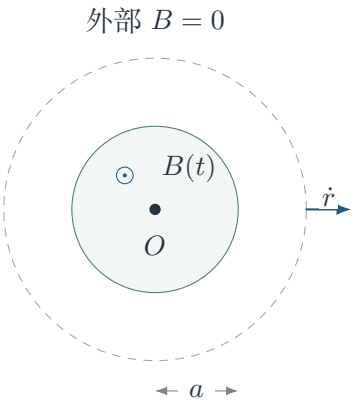


图 12-45　题图　同轴外圆环在无磁场区继续扩张

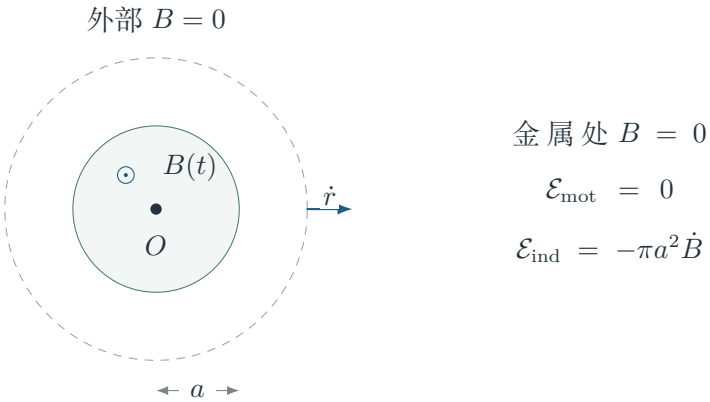


图 12-63　解答分析图　扩张扫过的区域无磁通，只有原磁通随外场改变

12-T9　径向非均匀场的电场分布

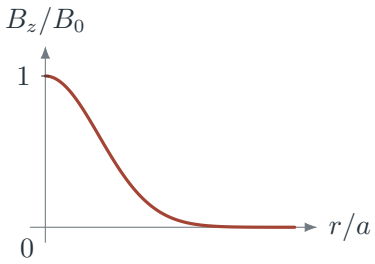


图 12-46　题图　高斯型径向场在轴心最大，向外衰减

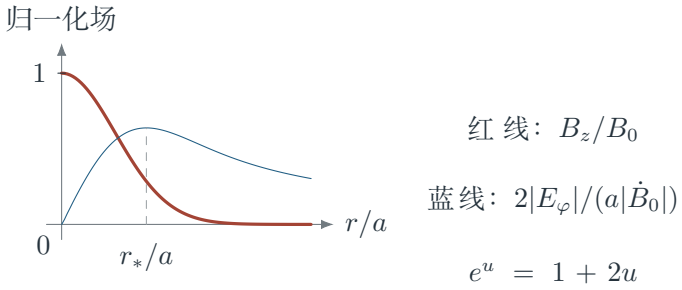


图 12-64　解答分析图　感应电场由圆内积分给出，在轴外达到最大值

12-T10　移动线圈的零电动势条件

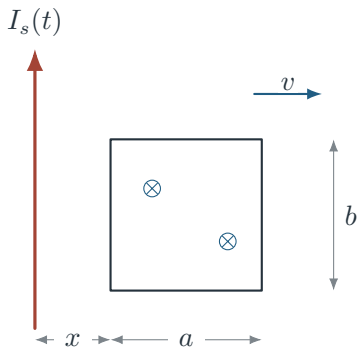


图 12-47　题图　线圈远离导线，同时调节原电流

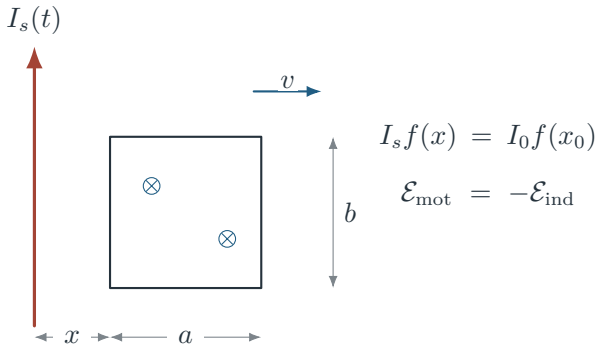


图 12-65　解答分析图　原电流增加恰好补偿几何因子的下降

12-T11　随时间变化的规范函数

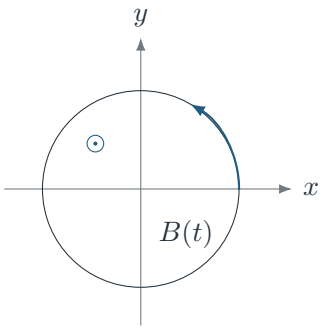
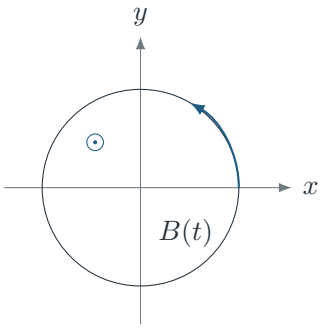


图 12-48　题图　逆时针固定圆周围住变化的均匀场



$$\chi = c(t)xy$$

$$\boldsymbol{A} \to \boldsymbol{A} + \nabla\chi$$

$$\phi \to \phi - \partial_t\chi$$

图 12-66　解答分析图　规范变换须同时改变标势，两种电场修正相消

12-T12　变化磁通对带电圆环的角冲量

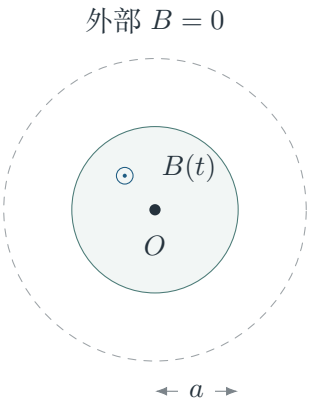
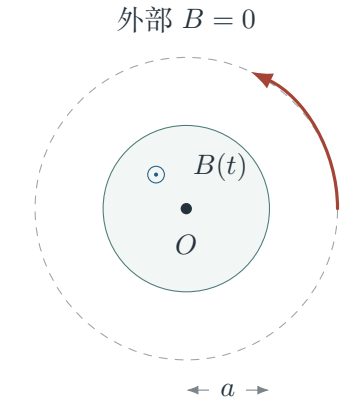


图 12-49　题图　带电绝缘圆环处于磁通区外，可绕轴自由转动



图取 $Q>0$, $\dot{\Phi}<0$

$J\omega_f = -Q\Delta\Phi/(2\pi)$

图 12-67　解答分析图　磁通下降时正电荷受逆时针电力，获得正角动量

12-T13　围住变化磁通的金属圆环片

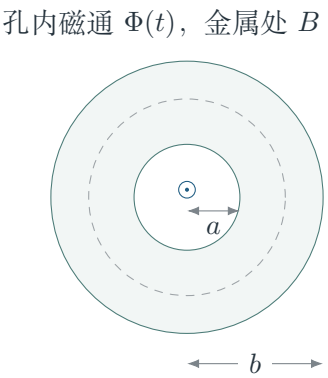
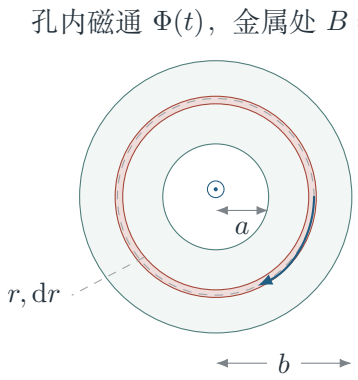


图 12-50　题图　薄金属环片围住变化磁通，外场不穿过金属



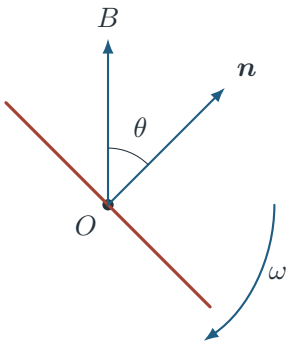
图取 $\dot{\Phi} > 0$

$E_\varphi = -\dot{\Phi}/(2\pi r)$

$P \propto \ln(b/a)$

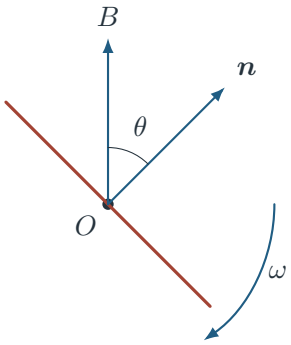
图 12-68　解答分析图　闭合环流仍非零，电流随半径按 $1/r$ 减小

12-T14　场振荡与线圈转动的频率



$$B = B_0 \cos \Omega t$$

图 12-51　题图　线圈转动与外场振荡具有不同角频率



$$B = B_0 \cos \Omega t$$

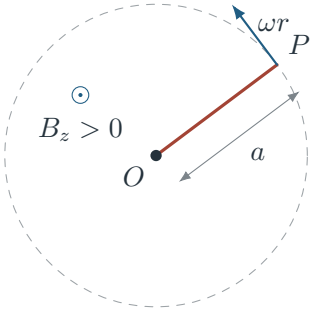
$$\Psi \propto \cos \Omega t \cos \omega t$$

频率为 $\Omega + \omega$

以及 $|\Omega - \omega|$

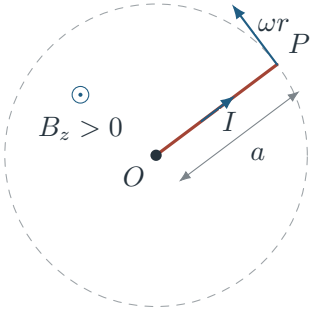
图 12-69　解答分析图　先展开磁通链，再求导，零频磁通不产生电动势

12-T15　边缘磁场为零的旋转棒



$$B_z(r) = B_0(1 - r^2/a^2)$$

图 12-52　题图　轴对称场在外端降为零，棒内部仍有非零磁场



$$B_z(r) = B_0(1 - r^2/a^2)$$

$$\mathcal{E} = \omega \int_0^a r B_z(r)\,\mathrm{d}r$$

$$\tau_B = -I \int_0^a r B_z(r)\,\mathrm{d}r$$

图 12-70　解答分析图　电动势与力矩都需沿全棒积分

12-T16　电容接入的自由滑动棒

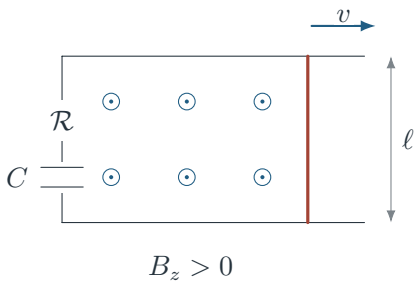


图 12-53　题图　左侧返回支路串接电阻与电容，棒自由向右滑动

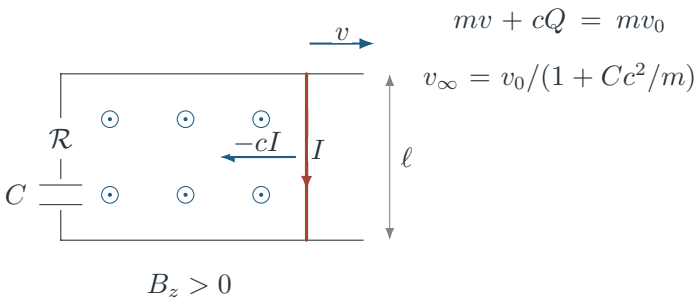


图 12-71　解答分析图　充电电流制动棒，最终电容电压与动生电动势平衡

12-T17　变电流与移动回路的能量（综合题）

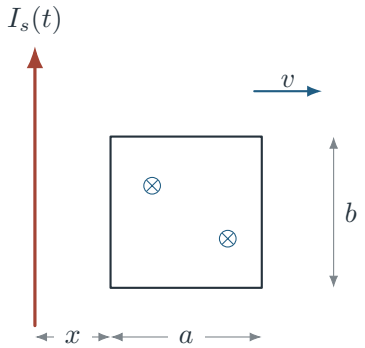


图 12-54　题图　原电流指数增长，线圈匀速离开直导线

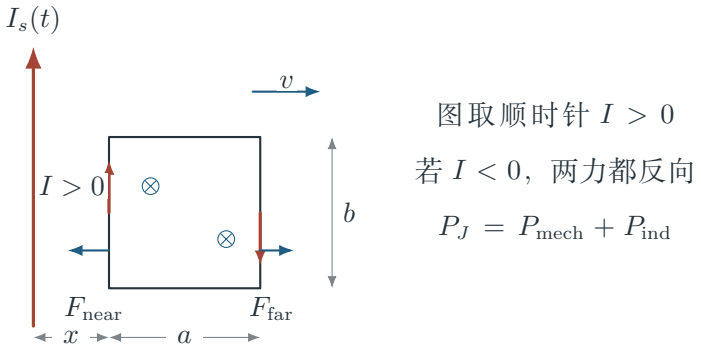


图 12-72　解答分析图　磁力方向依赖两个电动势的代数和，外场源可能帮助运动

12-T18　旋转棒与固定圆弧轨道（综合题）

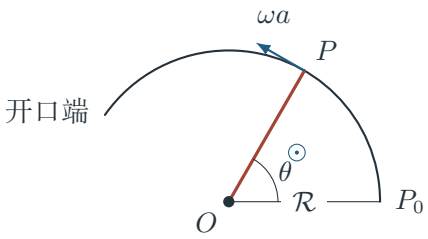
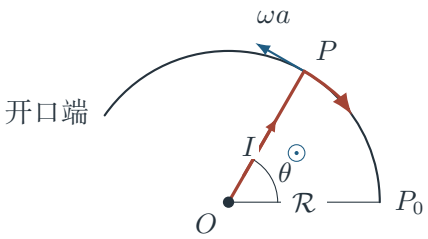


图 12-55　题图　旋转棒端在开口圆弧导轨上滑动，经固定径向电阻返回轴心



正回路沿棒向外

$\Phi = -Ba^2\theta/2$

$\mathcal{E} = c(B\omega + \beta\theta)$

图 12-73　解答分析图　正绕行方向为顺时针，棒中向外电流产生阻转矩